

DBMS EXAM GUIDE

Units 1, 2, and 5

Semi-Detailed Study Material

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UNIT 1: INTRODUCTION TO DBMS

1.1 Definition and Purpose of DBMS

A **Database Management System (DBMS)** is a software system that allows users to define, create, maintain, and control access to databases. It acts as an intermediary between the data and the application programs/users. A DBMS simplifies the complexity of data processing by providing tools to organize data, ensure integrity, and prevent unauthorized access or loss.

Key Functions:

Function	Description
Data Definition	Define the structure of database using DDL
Data Manipulation	Insert, update, delete, and retrieve data using DML
Data Control	Control access and provide security mechanisms
Data Integrity	Ensure consistency and accuracy of data
Concurrency Control	Manage multiple users accessing data simultaneously
Backup & Recovery	Protect against data loss and system failures

1.2 Advantages of DBMS

- Data Redundancy Control: Reduces duplicate data storage
- Data Consistency: Ensures data is accurate and up-to-date
- Data Security: Protects sensitive information from unauthorized access
- Concurrent Access: Multiple users can access data simultaneously
- Easy Data Access: Simple queries retrieve required information
- Backup and Recovery: Automatic backup mechanisms prevent data loss
- Data Integrity: Maintains accuracy through constraints and rules
- Standardization: Uniform data format across organization
- Query Processing: Efficient querying mechanisms
- Multiple Views: Different users see different data views

1.3 Users of DBMS

User Type	Role & Responsibilities
Database Administrator (DBA)	Manages database, ensures performance, security, backups
Database Designer	Designs database schema and structure
Application Programmers	Write backend code using languages like C, COBOL
End Users (Naive Users)	Use pre-written applications without knowing database details
Casual/Temporary Users	Occasionally access database for new information
Specialized Users	Write specialized applications (CAD systems, expert systems)

1.4 DBMS Architecture

Two-Tier Architecture (Client-Server): Direct communication between client and server with no intermediate layer. Client sends requests directly to database server.

Three-Tier Architecture: Client communicates with application server, which in turn communicates with database server. This provides better separation of concerns and scalability.

Three-Tier Layers:

Layer	Description	Purpose
Presentation (Client)	UI components, forms, displays	User interaction and input
Business Logic (Application)	Validation, calculations, data insertion logic	Data processing and rules
Data Access (Database)	Actual database with methods to connect and perform CRUD	Data storage and retrieval

1.5 Data Storage and Manager Components

Disk Storage Components:

- **Data Files:** Stores the actual database
- **Data Directory:** Contains metadata (data about data) and schema information
- **Indices:** Provides fast access to data items
- **Statistical Data:** Stores information and statistics about data

Storage Manager Components:

- **Authorization & Integrity Manager:** Checks user authority and data integrity constraints
- **Transaction Manager:** Ensures database consistency despite failures and manages concurrent execution
- **File Manager:** Manages disk space allocation
- **Buffer Manager:** Fetches data from disk to main memory and manages caching

Query Processor Components:

- **DDL Interpreter:** Interprets DDL statements and records definitions in data directory
- **DML Compiler:** Translates DML statements into low-level instructions; performs query optimization
- **Query Evaluation Engine:** Executes low-level instructions generated by DML compiler

UNIT 2: DATA MODELING AND ENTITY-RELATIONSHIP MODEL

2.1 Data Collection and Preprocessing

Data Collection is the process of gathering raw data from various sources. **Data Preprocessing** involves cleaning, transforming, and preparing data for analysis.

Key Preprocessing Steps:

- Data Cleaning: Remove or correct inconsistencies and errors
- Data Transformation: Convert data to appropriate formats
- Data Reduction: Remove irrelevant or redundant data
- Data Integration: Combine data from multiple sources
- Data Validation: Ensure data quality and consistency

2.2 Entity-Relationship (ER) Model

The **Entity-Relationship (ER) Model** is a conceptual data model used to represent data and relationships in databases. It provides a high-level design of the database independent of any specific DBMS. The ER model uses visual diagrams (ER diagrams) to show how data entities are related to each other.

2.3 ER Model Components

Entity: A thing of importance that can be distinguished, such as a person, place, or object. Represented as a rectangle.

Attribute: Properties or characteristics of an entity. - **Simple Attribute:** Cannot be divided further - **Composite Attribute:** Can be divided into smaller attributes - **Single-valued Attribute:** Has only one value - **Multi-valued Attribute:** Can have multiple values - **Derived Attribute:** Derived from other attributes

Relationship: Association between two or more entities. Represented as a diamond shape. Relationships have **cardinality** which defines how entities are related.

Cardinality Types:

Type	Notation	Meaning
One-to-One (1:1)	1 — 1	Each entity in set A relates to at most one entity in set B
One-to-Many (1:N)	1 — N	One entity in A can relate to many entities in B
Many-to-One (M:1)	M — 1	Many entities in A relate to one entity in B
Many-to-Many (M:N)	M — N	Many entities in A relate to many entities in B

Participation Constraint: Specifies whether every entity participates in a relationship.

- **Total Participation:** Every entity must participate in the relationship - **Partial Participation:** Not all entities need to participate in the relationship

Key in ER Model:

A **key** is a set of attributes that uniquely identifies an entity instance. - **Primary Key:** The chosen unique identifier - **Candidate Key:** Any attribute(s) that can uniquely identify an entity - **Composite Key:** A key made up of multiple attributes

UNIT 5: RELATIONAL MODEL

5.1 Introduction to Relational Model

The **Relational Model** organizes data using tables (relations) consisting of rows and columns. It transforms conceptual designs from ER diagrams into implementable structures used in relational databases like Oracle SQL and MySQL. Each row represents an entity, each column represents an attribute.

5.2 Key Terms in the Relational Model

Term	Definition
Attribute	A property that defines an entity (e.g., NAME, ROLL_NO)
Relation Schema	Structure of relation showing name and attributes (e.g., STUDENT(ROLL_NO, NAME, ADDRESS))
Tuple	A row in a relation representing a single entity instance
Relation Instance	Set of tuples at a particular point in time; changes with insertions/deletions
Degree	Number of attributes in a relation
Cardinality	Number of tuples in a relation
NULL Value	A value that is unknown or unavailable

5.3 Types of Keys in Relational Model

Key Type	Definition
Primary Key	Uniquely identifies each tuple; must be unique, cannot be NULL
Candidate Key	Any attribute(s) that can uniquely identify a tuple
Super Key	Set of attributes that can uniquely identify a tuple
Foreign Key	Attribute in one relation that refers to primary key of another relation
Composite Key	Two or more attributes combined to uniquely identify a tuple

5.4 Constraints in Relational Model

Domain Constraints: Each attribute value must be atomic from its defined domain (data type). Common types:

- Numeric: Integers and real numbers
- Character: Fixed-length (CHAR) or variable-length (VARCHAR) strings
- Boolean: True or false values
- Specialized: DATE, TIME, TIMESTAMP, MONEY

Key Integrity Constraint: Every relation must have a primary key with two properties:

- Must be unique for all tuples
- Cannot contain NULL values

Referential Integrity Constraint: A foreign key value must either match a primary key value in the referenced relation or be NULL.

5.5 Anomalies in Database Operations

Insertion Anomaly: Cannot insert a row in a referencing relation if the foreign key value does not exist in the referenced relation.

Deletion Anomaly: Cannot delete a row from a referenced relation if its primary key is still referenced by foreign keys in other relations.

Update Anomaly: Cannot update a primary key value if it is referenced by foreign keys in other relations.

5.6 SQL Functions

Aggregate Functions: Operate on multiple values and return a single result.

- COUNT(): Returns number of rows
- SUM(): Returns sum of numeric values
- AVG(): Returns average of numeric values
- MAX(): Returns maximum value
- MIN(): Returns minimum value
- FIRST(): Returns first value in result set
- LAST(): Returns last value in result set

Scalar Functions: Operate on single values and return single value.

- UCASE(): Convert string to uppercase
- LCASE(): Convert string to lowercase
- MID(): Extract substring from string
- LEN(): Return string length
- ROUND(): Round numeric value
- NOW(): Return current date/time
- FORMAT(): Format date/number in specific format

5.7 Characteristics of Relational Model

- Data organized in tables (relations) with rows (tuples) and columns (attributes)
- Atomic values: Each cell contains single, indivisible value
- Unique keys: Every table has primary key ensuring no duplicates
- Attribute domain: Each attribute has defined data type and constraints
- Data independence: Changes to schema don't affect applications
- Relational operations: Supports selection, projection, join, union, intersection
- Data consistency: Constraints reduce redundancy and anomalies
- Set-based representation: Operations follow mathematical set theory principles

KEY POINTS SUMMARY FOR EXAM

UNIT 1 - DBMS Fundamentals:

- DBMS provides data definition, manipulation, control, integrity, concurrency control, and backup/recovery
- DBA, designers, programmers, and users interact with DBMS differently
- Architecture supports 2-tier (client-server) and 3-tier (presentation-business-data) models
- Storage manager handles authorization, transactions, file management, and buffering
- Query processor includes DDL interpreter, DML compiler, and query evaluation engine

UNIT 2 - Data Modeling:

- ER model uses entities, attributes, and relationships to model data
- Cardinality defines relationship types: 1:1, 1:N, M:1, M:N
- Participation constraints are total (mandatory) or partial (optional)
- Data preprocessing includes cleaning, transformation, reduction, integration, validation
- ER diagrams provide conceptual design independent of specific DBMS

UNIT 5 - Relational Model:

- Relational model converts ER diagrams into implementable table structures
- Primary keys uniquely identify tuples; foreign keys reference other relations
- Three types of constraints: domain, key integrity, and referential integrity
- Anomalies: insertion (cannot insert with non-existent foreign key), deletion/update (cannot modify referenced values)
- SQL provides aggregate functions (COUNT, SUM, AVG, MAX, MIN) and scalar functions (UCASE, LCASE, MID, etc.)
- Relational operations: selection, projection, join, union, intersection follow set theory